

FITCORE PRO

Unified Fitness Industry Platform

COMPREHENSIVE PRODUCT BLUEPRINT

Architecture · Strategy · Roadmap

Version 1.0 · June 2026

1. Executive Summary

FitCore Pro is a next-generation, multi-sided SaaS platform engineered to become the operating system of the fitness industry. Unlike Fitboat (a gym-management SaaS limited to scheduling and billing) or GYM-One (a self-hosted open-source tool with no marketplace layer), FitCore Pro connects every actor in the fitness ecosystem — gyms, trainers, nutritionists, supplement companies, equipment manufacturers, maintenance providers, and end-customers — on a single, AI-powered, revenue-sharing platform.

1.1 The Gap FitCore Pro Fills

Capability	Fitboat	GYM-One	FitCore Pro
Multi-role marketplace	X	X	
AI recommendations	X	X	
OYO-style pay-per-use	X	X	
Local SEO & discovery	Basic	X	
Biometric scanning	X	X	
Supplement marketplace	X	X	
Equipment lifecycle mgmt	X	X	
Maintenance scheduling	X	X	
Revenue sharing engine	X	X	
Mobile-first + Web	Partial	X	

1.2 Business Model Summary

FitCore Pro earns revenue across four streams:

- SaaS subscriptions — tiered plans for gyms, trainers, and companies
- Marketplace commission — 3-8% on supplement, equipment, and PT session bookings
- Lead generation fee — charged to gyms/trainers for verified customer leads
- Data intelligence — anonymised benchmarks sold to supplement and equipment companies

1.3 Market Opportunity

The global gym management software market is projected to exceed \$5.6 billion by 2030 (CAGR ~9.4%). India alone has 300,000+ gyms, of which fewer than 12% use any software. FitCore Pro

targets this underserved majority while simultaneously building a premium upsell path — identical to how OYO disrupted the unbranded hotel segment.

2. Feature List by Role (Summary)

2.1 Gym Owners / Management

- Multi-branch management with unified dashboard and real-time analytics
- Member lifecycle automation: onboarding → renewal → churn prediction
- Trainer roster management with PT commission tracking and scheduling
- Supplement product showcase and zero-inventory consignment display
- Automated renewal reminders via SMS, WhatsApp, push, and email
- Biometric device integration (fingerprint, face, QR, NFC)
- White-label mobile app branding and Google My Business sync

2.2 Trainers (In-Gym & Independent)

- Professional profile with certifications, ratings, and discovery in marketplace
- PT session scheduling with calendar sync and automated reminders
- Client portfolio with biometric tracking and progress history
- Supplement recommendation engine with affiliate commission tracking
- Earnings dashboard: PT fees, supplement commissions, performance bonuses

2.3 Customers / End-Users

- Goal-based onboarding: yoga, zumba, powerlifting, bodybuilding, weight loss, rehab
- Gym and trainer discovery with GPS, filters, ratings, and booking
- Flexible membership: fixed monthly, pay-per-visit, class bundles with no upfront commitment
- Biometric gym entry: QR, NFC, facial recognition — no card needed
- Workout and progress tracker with AI-generated plans based on biometrics and goals
- Diet and nutrition log with macro calculator and meal recommendations
- Supplement shop with trainer recommendations, reviews, and delivery
- Auto-renewal with smart alerts and one-tap renewal functionality
- Loyalty points, referral rewards, and milestone badges

2.4 Nutritionists, Supplement Companies & Equipment Manufacturers

Nutritionists:

- Client management, diet plan builder, consultation scheduling, discovery profile

Supplement Companies:

- Brand account with zero-inventory virtual shelf presence in gyms, sales analytics by gym/region

Equipment Manufacturers:

- Equipment catalogue, installation requests, AMC lifecycle, maintenance job dispatch, machine telemetry integration

2.5 Maintenance Providers & Company Staff

Maintenance:

- Job board filtered by location and skill, en-route tracking, job card completion, earnings dashboard

Company Staff (Area Managers, Field Officers):

- Territory dashboard, gym visit logs, NPS surveys, revenue tracking, onboarding pipeline, support ticket management

3. System Architecture

3.1 Architecture Pattern: Modular Monolith → Microservices

FitCore Pro adopts a phased architecture strategy:

- Phase 1-2: Modular Monolith with clear domain boundaries (faster to build, easier to debug)
- Phase 3+: Extract hot modules to microservices as scale demands (Payments, AI, Notifications, Search)
- All modules communicate via internal service interfaces first, then gRPC/REST for extracted services

3.2 High-Level System Components

CLIENT TIER React Native mobile app (iOS + Android), React.js Progressive Web App, Admin portal (React.js), Biometric devices (SDK bridge)

API GATEWAY Kong or AWS API Gateway — rate limiting, auth, routing, SSL termination, API versioning

CORE SERVICES User & Identity · Gym Management · Booking & Scheduling · Payments · Notifications · Search & Discovery · AI/ML Engine · Supplement Marketplace · Equipment & Maintenance · Analytics

DATA TIER PostgreSQL (primary relational) · Redis (cache + session) · Elasticsearch (search) · ClickHouse (analytics) · S3/Cloudflare R2 (media)

INFRA & OPS Kubernetes (EKS/GKE) · ArgoCD (GitOps) · Prometheus + Grafana · Sentry · CloudFlare CDN

3.3 Core Database Tables (Schema Overview)

Multi-tenancy is implemented using row-level security (RLS) with tenant_id column on all tables.

users id, tenant_id, role_id, email, phone, password_hash, first_name, last_name, avatar_url, status, created_at

gyms id, owner_id, name, slug, address, city, state, lat, lng, phone, logo_url, gmb_place_id, subscription_plan_id, status

memberships id, gym_id, customer_id, plan_id, start_date, end_date, status, payment_method, price_paid, auto_renew, renewal_date

membership_plans id, gym_id, name, type (fixed|payg|flex), duration_days, price, sessions_included, features (JSONB)

trainers id, user_id, gym_id (nullable), bio, certifications (JSONB), specialisations[], hourly_rate, is_independent, rating, total_reviews

pt_sessions id, trainer_id, customer_id, gym_id, scheduled_at, duration_mins, status, notes, payment_id

workouts id, customer_id, trainer_id (nullable), gym_id, date, exercises (JSONB array), duration_mins, ai_generated

body_metrics id, user_id, recorded_at, weight_kg, body_fat_pct, muscle_mass_kg, bmi, measurements (JSONB), photos[]

diet_plans id, nutritionist_id, customer_id, name, start_date, end_date, daily_calories, macros (JSONB), meals (JSONB array)

supplements id, company_id, name, sku, category, description, images[], price, stock_type (dropship|physical), status

equipment id, manufacturer_id, gym_id, model, serial_number, purchase_date, warranty_expiry, status, last_serviced_at

maintenance_jobs id, equipment_id, gym_id, technician_id, reported_at, scheduled_at, completed_at, status, notes, photos[]

payments id, entity_type, entity_id, amount, currency, gateway, gateway_txn_id, status, refund_amount, metadata (JSONB)

notifications id, user_id, channel (sms|email|push|whatsapp), type, payload (JSONB), scheduled_at, sent_at, status

4. Technology Stack

Layer	Technology	Rationale
Mobile App	React Native + Expo	Single codebase for iOS & Android; OTA updates
Web Frontend	Next.js 14 (React)	SSR for SEO; ISR for gym listing pages
Admin Portal	React.js + Vite	Fast, composable, real-time dashboards
API Backend	Node.js (NestJS)	TypeScript, modular, DI, OpenAPI auto-docs
AI/ML Service	Python (FastAPI)	ML model serving; separate compute scaling
Primary Database	PostgreSQL 16	ACID compliance; multi-tenant RLS
Cache	Redis 7 (Cluster)	Sub-ms latency; pub/sub for notifications
Search Engine	Elasticsearch 8	Gym discovery, full-text, geosearch
Analytics DB	ClickHouse	Columnar; billion-row queries in seconds
Message Queue	BullMQ (Redis)	Job queues for emails, reminders, AI jobs
File Storage	Cloudflare R2	S3-compatible; no egress fees
Payments	Razorpay + Stripe	UPI, cards, wallets, EMI, international
SMS / WhatsApp	Twilio / Gupshup	OTP, reminders, renewal alerts
Push Notifications	Firebase Cloud Messaging	iOS + Android; Expo push proxy
Maps	Google Maps Platform	Places API, distance matrix, GMB sync
Video Calls	Agora.io / Jitsi	PT and nutritionist consultations
Containerisation	Docker + Kubernetes	Auto-scaling; zero-downtime deploys
CI/CD	GitHub Actions + ArgoCD	GitOps; preview environments
Monitoring	Prometheus + Grafana	Metrics, traces, error tracking (Sentry)

5. Development Roadmap & Phases

Phase	Timeline	Deliverables	Team Size
0	Weeks 1-4	Design & Discovery: Wireframes for all 8 roles, DB schema, API contracts, design system, infrastructure setup	
1	Months 2-4	MVP Core: Auth, Gym onboarding, Memberships, Trainer profiles, Basic booking, Razorpay, SMS reminders, Mobile app	
2	Months 5-7	Marketplace: Supplements, Equipment, Maintenance jobs, Biometric device integration, Push notifications, QR entry	
3	Months 8-10	AI & Intelligence: Churn prediction, AI workouts/diets, Offers, ClickHouse analytics, Local SEO, GMB sync	
4	Months 11-12	Scale & Polish: White-label app, Franchise tools, Video calls, Wearable sync, Multi-language, Hardening	
5	Month 13+	Expansion: International payments, IoT telemetry, B2B data intelligence, Partner APIs	

5.1 Recommended Initial Team (Phase 0-1)

- 1× Product Manager — define user stories, roadmap prioritisation
- 1× UI/UX Designer — wireframes, prototypes, design system
- 2× Backend Engineers (Node.js/NestJS) — API layer, DB design
- 2× Frontend Engineers (React.js + React Native) — web and mobile
- 1× DevOps / Infrastructure Engineer — Docker, Kubernetes, CI/CD
- 1× QA Engineer — automated testing (unit, integration, E2E)

6. Risk Mitigation Strategy

Risk	Severity	Likelihood	Mitigation
Data breach / DPDP violation	Critical	Medium	
Payment gateway downtime	High	Low	
Biometric vendor lock-in	Medium	Medium	
Large gym churn	High	Low	
AI recommendation quality	Medium	Medium	
Supplement regulatory issues	High	Medium	
Scalability at 1M+ users	High	Low	
Field sales performance	Medium	Medium	
Off-platform trainer deals	Medium	High	
Multi-state GST compliance	Medium	High	

6.1 Security & Compliance Checklist

- OAuth 2.0 + JWT (RS256) with 7-day refresh token rotation
- Rate limiting at API gateway (per IP, per API key)
- Input validation (Zod) on every endpoint
- SQL injection prevention: parameterised queries only
- OWASP Top 10 review at major releases
- PCI-DSS delegated to Razorpay/Stripe (no raw card storage)
- Biometric data stored as one-way templates, never raw images
- GDPR / India DPDP Act 2023: consent management, right to erasure, data portability
- Weekly security scans: Snyk (dependencies), OWASP ZAP (API)

7. Immediate Next Steps (Week 1-2)

1. Lock MVP scope: select 2 pilot cities and 3 gym partners for validation
2. Hire/contract core team: NestJS backend lead, React Native engineer, UI/UX designer
3. Setup infrastructure: GitHub org, AWS account, Razorpay test sandbox, Figma
4. Begin wireframing: Gym dashboard, Customer home, Trainer schedule, Supplement shop
5. Draft partnership agreement: data sharing, commission structure, white-label terms

7.1 Success Metrics (First 90 Days)

- Pilot gyms onboarded: 10 target
- Active members using app: 500 target
- QR / biometric entry events: 5,000 logged (validates hardware)
- Supplement orders: 100 placed (marketplace validation)
- App store rating: 4.2+ (UX quality validation)

7.2 What Makes FitCore Pro Defensible

- Network effects: more gyms → more customers → more trainers → more data → better AI → stronger moat
- Switching costs: biometric hardware, member history, trainer schedules, supplement relationships = lock-in
- Data moat: workout, biometric, nutrition data at scale enables AI no single-player tool can match
- Revenue alignment: platform earns when gyms earn — genuine partnership model, not extraction

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